

Decent Coaching Classes

X SB

Mind Your Math - VII

Marks: 40

Time: 2 Hrs

Instructions: All questions are compulsory (internal choice as indicated). Figures to the right indicate full marks. Use of calculator is not allowed.

Q.1 (A) Choose the correct alternative. [4]

- (1) For the simultaneous equations $2x + 3y = 5$ and $4x - y = 3$, the value of D is _____.
(A) -14 (B) 14 (C) -2 (D) 10
- (2) The nature of the roots of the equation $x^2 - 4x + 4 = 0$ is _____.
(A) real and equal (B) real and unequal (C) not real (D) cannot be said
- (3) The 15th term of the A.P. 4, 7, 10, ... is _____.
(A) 42 (B) 46 (C) 49 (D) 45
- (4) A die is rolled once. The probability of getting a prime number is _____.
(A) $1/2$ (B) $1/3$ (C) $2/3$ (D) $1/6$

Q.1 (B) Solve the following. [4]

- (1) Write the formula for the sum of the first n terms of an A.P. in terms of the first term a and the last term t_n .
- (2) Write the sample space when two coins are tossed simultaneously.
- (3) The total value of a mutual fund scheme is ₹300 crore and 10 crore units are issued. Find the NAV of one unit.
- (4) Find the class mark of the class 150 – 200.

Q.2 (A) Complete any two of the following activities. [4]

- (1) Solve the following simultaneous equations by Cramer's rule: $x + y = 3$; $3x - 2y = 4$
 $D = 1 \times (-2) - 1 \times 3 = \underline{\hspace{2cm}}$, $D_x = 3 \times (-2) - 1 \times 4 = \underline{\hspace{2cm}}$, $D_y = 1 \times 4 - 3 \times 3 = \underline{\hspace{2cm}}$
 $x = D_x/D = \underline{\hspace{2cm}}$, $y = D_y/D = \underline{\hspace{2cm}}$
- (2) Solve by factorisation: $m^2 - 14m + 13 = 0$
 $m^2 - \underline{\hspace{2cm}} - m + 13 = 0$
 $m(m - \underline{\hspace{2cm}}) - 1(m - \underline{\hspace{2cm}}) = 0$
 $(m - 13)(\underline{\hspace{2cm}}) = 0$, $\therefore m = \underline{\hspace{2cm}}$ or $m = \underline{\hspace{2cm}}$
- (3) Find the sum of the first 20 terms of the A.P. 5, 9, 13, ...
Here $a = 5$, $d = \underline{\hspace{2cm}}$, $n = 20$. $S_n = n/2 [2a + (n - 1)d]$
 $\therefore S_{20} = \underline{\hspace{2cm}} [2 \times 5 + \underline{\hspace{2cm}} \times 4] = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

Q.2 (B) Solve any two of the following. [4]

- (1) Solve by elimination method: $2x + 3y = 12$; $x - y = 1$
- (2) Find the discriminant of $2x^2 + 4x + 5 = 0$ and hence write the nature of its roots.
- (3) One card is drawn at random from a well shuffled pack of 52 playing cards. Find the probability that the card is (i) a king (ii) a red card.

Q.3 (A) Complete any one of the following activities.**[3]**

- (1) The taxable value of goods is ₹5000 and the rate of GST is 12%.

Rate of CGST = _____ %, Rate of SGST = _____ %

CGST = ₹ _____, SGST = ₹ _____, Total GST = ₹ _____

Invoice value = 5000 + _____ = ₹ _____

- (2) Find the mean of the following data:

Class: 0 – 10 10 – 20 20 – 30 30 – 40 40 – 50

Frequency (f_i): 5 8 15 16 6

Class marks (x_i) = 5, 15, _____, 35, _____; Σf_i = _____; $\Sigma f_i x_i$ = 25 + 120 + _____ + 560 + 270 = _____

Mean = $\Sigma f_i x_i / \Sigma f_i$ = _____ / _____ = _____

Q.3 (B) Solve any two of the following.**[6]**

- (1) Solve using the formula: $2x^2 - 7x + 3 = 0$
- (2) How many terms are there in the A.P. 7, 13, 19, ..., 205?
- (3) Two dice are rolled simultaneously. Write $n(S)$ and find the probability that the sum of the numbers on the upper faces is 9.

Q.4 Solve any three of the following.**[12]**

- (1) The sum of the digits of a two digit number is 9. The number obtained by interchanging the digits exceeds the original number by 27. Find the original number.
- (2) The sum of the squares of two consecutive natural numbers is 113. Find the numbers.
- (3) A dealer purchased goods worth ₹50,000 and sold them for ₹70,000, GST being 12% on both. Find the input tax credit, the GST payable by the dealer, and the CGST and SGST payable.
- (4) Find the mean of the following data by the assumed mean method:
- Class: 10 – 20 20 – 30 30 – 40 40 – 50 50 – 60
- Frequency: 4 10 16 8 2
- (5) A person saves ₹1000 in the first month and increases his saving by ₹200 every month. Find his total saving in 12 months.

Q.5 Solve any one of the following.**[3]**

- (1) Mr. Joshi bought 40 shares of face value ₹100 at the market value of ₹150 each, paying brokerage at 0.5%. Find (i) the total amount he paid (ii) the dividend he receives if the company declares a dividend of 12%.
- (2) Ms. Kale invested ₹6000 in a mutual fund scheme when the NAV was ₹24 per unit. After one year the NAV is ₹28.50. Find the number of units allotted and the gain on her investment.